

CO5-AT2-Application Based Problem Solving

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QUESTIONS

Q1

Question 2 — Hospital Appointment Management
100 marks

1/1 answered

Q1 Question 2 — Hospital Appointment Management

100 marks

Design a class hierarchy with a base class Staff, and derived classes Doctor and Nurse. Use overloaded constructors to initialize staff with/without a specialization. Overload a method assignDuty() for single vs. multiple patients. Override a generateReport() method differently in Doctor and Nurse. Create a custom checked exception AppointmentConflictException thrown when two patients are booked for the same doctor at the same time. Use multiple threads to represent patients booking appointments concurrently with a single doctor's schedule.

Evaluate: Critically assess what happens if AppointmentConflictException is checked vs. unchecked in this context. Justify your choice of exception type and your synchronization approach for the shared schedule, considering both correctness and system responsiveness (avoiding long lock hold times).

Your Code

```
1 class Main {
2     public static void main(String[] args) {
3         Doctor d = new Doctor("Dr. Ravi", "Cardiology");
4         Nurse n = new Nurse("Nurse Anu");
5         d.assignDuty("Patient A");
6         d.assignDuty("Patient B", "Patient C");
7         n.assignDuty("Patient D");
8         System.out.println(d.generateReport());
9         System.out.println(n.generateReport());
10        Thread t1 = new Thread(() -> book(d, "10:00", "Patient A"));
11        Thread t2 = new Thread(() -> book(d, "10:00", "Patient B"));
12        t1.start();
13        t2.start();
14    }
15    static void book(Doctor d, String time, String patient) {
16        try {
17            d.book(time, patient);
18        } catch (AppointmentConflictException e) {
```

Input

stdin input

Output

```
Dr. Ravi assigned to Patient A
Dr. Ravi assigned to Patient B
and Patient C
Nurse Anu assigned to Patient D
Doctor Report: Dr. Ravi -
Cardiology
```